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Lancet Diabetes Endocrinol. Author manuscript; available in PMC 2022 September 01.

Published in final edited form as:
Lancet Diabetes Endocrinol. 2021 September ; 9(9): 622–637. doi:10.1016/S2213-8587(21)00173-X.

Bone Mineral Density Surveillance for Childhood, Adolescent, and Young Adult Cancer Survivors: Recommendations from the International Late Effects of Childhood Cancer Guideline Harmonization Group

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Summary

Childhood, adolescent and young adult (CAYA) cancer survivors are at increased risk of reduced bone mineral density (BMD). Clinical practice surveillance guidelines are important for timely diagnosis and treatment of these survivors, which could improve BMD parameters and prevent fragility fractures. Discordances across current late-effects guidelines necessitated international harmonization of recommendations for BMD surveillance. The International Late Effects of Childhood Cancer Guideline Harmonization Group therefore established a panel of 36 experts from 10 countries, representing a range of relevant medical specialties. The evidence of risk factors for (very) low BMD and fractures, surveillance modality, timing of BMD surveillance, and treatment of (very) low BMD were evaluated and critically appraised, and harmonized recommendations for CAYA cancer survivors were formulated. We graded the recommendations based on the quality of evidence and balance between potential benefits and harms. BMD surveillance is recommended for survivors treated with cranial or craniospinal radiotherapy, and is reasonable for survivors treated with total body irradiation. Due to insufficient evidence, no recommendation can be formulated for or against BMD surveillance for survivors treated with corticosteroids. This surveillance decision should be made by the survivor and healthcare provider together, after careful consideration of the potential harms and benefits and additional risk factors. We recommend to perform BMD surveillance using dual-energy X-ray absorptiometry at entry into long-term follow-up, and if normal (Z-score >−1), repeat at 25 years of age. Between these measurements and thereafter, surveillance should be performed as clinically indicated. These recommendations facilitate evidence-based care for CAYA cancer survivors internationally.

Introduction

The survival of children, adolescents and young adults (CAYAs) with cancer has greatly improved over recent decades, with current 5-year overall survival rates approximating 80% in high-income countries.[1,2] However, CAYA cancer survivors often experience long-term side effects.[3,4] Previous studies suggest an increased proportion of survivors with

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low (Z-score ≤−1) and very low (Z-score ≤−2) bone mineral density (BMD) compared to the normal population,[5,6] as well as an increased fracture rate.[7,8] BMD deficits may occur due to the cancer itself, its treatment, or consequences such as endocrine defects (e.g. hypogonadism, growth hormone deficiency [GHD]), malnutrition or malabsorption, and sedentary lifestyles.[9–11] These factors may lead to impaired bone accrual, resulting in a lower peak bone mass (PBM), usually achieved between the age of 20–30 years.[12,13] Because PBM predicts osteoporosis in adulthood and affects the age of osteoporosis onset,[14] it is hypothesized that as the current CAYA cancer survivor population ages, more survivors may experience fragility fractures at relatively young ages.[15] These fragility fractures may cause substantial morbidity such as reduced mobility, chronic pain, and difficulty with activities of daily living.[16]

General population studies have shown that early detection and treatment of (very) low BMD may overcome suboptimal PBM acquisition and prevent fractures.[17,18] Therefore, several North American and European groups have implemented BMD surveillance in their clinical practice survivorship guidelines.[19–22] However, these guidelines have not systematically analyzed the literature. Thus, definitions of high-risk groups, timing of surveillance, and treatment recommendations vary considerably, which impedes effective international implementation and adherence. To overcome such limitations, the International Late Effects of Childhood Cancer Guideline Harmonization Group (IGHG) was established.[23] This collaborative endeavor aims to establish a common vision and integrated strategy for surveillance of chronic health problems in CAYA cancer survivors. This IGHG report summarizes available evidence and provides the first harmonized recommendations for BMD surveillance among CAYA cancer survivors.

Methods

Guideline panel

The guideline panel was composed of 36 experts from 10 countries, representing several pediatric oncology and other related societies, as well as a broad range of medical specialties (Appendix pp. 2–3). Three dedicated working groups addressed the following topics (Appendix p. 4): 1) who needs BMD surveillance?; 2a) what surveillance modality should be used?; 2b) when should surveillance be initiated and at what frequency should it be performed?; and 3) what should be done when abnormalities are identified? Details on the IGHG methodology have been previously reported (Appendix p. 5).[23]

Scope and definitions

This guideline provides BMD surveillance recommendations for CAYA cancer survivors diagnosed with cancer up to 25 years of age, who are at least two years after completion of treatment, regardless of current age. Definitions of the main potential osteotoxic treatments and outcomes can be found in Appendix p. 6. We analyzed risk factors for very low BMD (including studies with a Z-score ≤−2 as outcome), low BMD (Z-score ≤−1 and ≤−2), lower BMD Z-score (continuous), and for fractures (all types). A Z-score indicates the number of standard deviations that BMD differs from age- and sex-matched normative values, whereas a T-score compares BMD with the healthy young adult mean (PBM). Z-scores (and not

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T-scores) were used as all included studies were performed in CAYAs, who may not have yet achieved PBM.[24,25] Although we mainly focused on risk factors for very low BMD, we also included studies using a BMD Z-score threshold ≤−1. This threshold was considered relevant in the context of CAYA BMD surveillance, because Z-scores ≤−1 but >−2 may predispose to developing very low BMD as survivors age, and because adult studies showed a 2–3 times increased fracture risk for every one standard deviation reduction in BMD.[26] All skeletal sites were analyzed together, because risks of (very) low BMD for different skeletal sites have similar implications for BMD surveillance.

Search strategy and selection criteria

Initially, the panel evaluated concordances and discordances between the online available Children's Oncology Group, Dutch Childhood Oncology Group, UK Children's Cancer and Leukaemia Group, and Scottish Intercollegiate Guideline Network survivorship guidelines.[19–22] Clinical questions were then formulated for all discordant areas. For concordant areas, clinical questions were drafted if there was any uncertainty about sufficient literature support or if more details were needed.

We searched English literature published between January 1, 1990 and May 12, 2021 using the MEDLINE (PubMed) database to identify relevant evidence (Appendix p. 13). Two independent reviewers determined whether the identified articles met pre-determined inclusion criteria (Appendix p. 15). Subsequently, all guideline members were contacted to determine if additional evidence was available. Cross-references identified during the review procedure that had not been initially identified were added if relevant.

We summarized information from included studies using evidence tables, and generated a summary of the total body of evidence per clinical question. The quality of the evidence was graded using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) methodology (Appendix pp. 16–20).[27] If evidence in CAYA cancer survivors was not available or insufficient to answer a clinical question, we searched for recommendations in clinical practice guidelines on osteoporosis or low BMD in closely related populations (Appendix p. 14). These recommendations were extrapolated to support expert opinion after careful consideration.

Translating evidence into recommendations

We used the GRADE Evidence-to-Decision framework to formulate recommendations.[28] Our harmonized recommendations for BMD surveillance were based on evidence, expert opinions, cost-benefit considerations, the balance between potential benefits and harms of surveillance, and the need to maintain flexibility of application across different health-care systems. According to the IGHG methodology, only treatment-related risk groups were considered for BMD surveillance. Decisions were made through group discussion and consensus. The strength of the recommendations were graded according to published evidence-based methods (Appendix p. 21).[27,29] Finally, two independent experts in the field (CS and J-MK) and six survivor representatives (DC, LG, MB, TC, MS, and ZT) had the opportunity to provide input on the harmonized recommendations.

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Results

Appendix pp. 7–8 shows the evaluation of concordant and discordant areas between the existing survivorship guidelines. Because the panel felt that both concordant and discordant areas required systematic, in-depth review of the evidence, clinical questions were drafted for all areas (Appendix pp. 9–12). Seventy-four studies in CAYA cancer survivors (Figure 1) and three clinical practice guidelines in related populations (two childhood cancer guidelines and one general pediatric guideline, Appendix pp. 22–26) were included. Evidence tables and summary of findings tables are presented in Appendix pp. 27–218 and pp. 219–365.

Risk of reduced BMD and fractures

CAYA cancer survivors are at increased risk of low BMD (moderate-quality evidence),[6,7,9–11,30–70] very low BMD (moderate-quality evidence),[6,7,9,10,30–37,39–44,47–55,58–62,64,66–71] and lower BMD Z-scores (moderate-quality evidence) after a follow-up period ranging from 2·7 to 27·2 years.[30,31,33,38,39,41,43,44,50,52–55,58,60,61,64,66,67,69–79] We found an increased risk of fractures in survivors versus controls (very low-quality evidence) after a follow-up ranging from 9·1 to 22·7 years.[7,8,80] (See Appendix pp. 219–237 for details on prevalence of (very) low BMD and incidence of fractures). Clinical fractures are significantly associated with low BMD in survivors (low-quality evidence),[58,68] but not with lower versus higher BMD Z-score as a continuum (moderate-quality evidence).[81] It is unknown whether (very) low BMD during therapy or previous fractures lead to an increased risk of reduced BMD and fractures at two or more years after treatment cessation (no studies).

Risk factors for very low BMD

Identified risk factors associated with very low BMD include cranial or craniospinal radiotherapy (C[S]RT) (high-quality evidence),[6,54] abdominal/pelvic irradiation (moderate-quality evidence, one study),[6] hypogonadism (moderate-quality evidence),[10,54] GHD (low-quality evidence),[10,54,82,83] low body mass index (BMI) (high-quality evidence),[6,54,69] and male sex (moderate-quality evidence).[6,41,54] For the remaining potential risk factors, no increased risk of very low BMD was found, or no or only very low-quality evidence was available (Table 1).

Treatment-related risk factors for low BMD

The main treatment-related risk factors for low BMD include C(S)RT (high-quality evidence),[6,9,32,38,51,54,64,68] total body irradiation (TBI) (high-quality evidence),[9,45,54,64,65] and corticosteroids (administered as anti-cancer treatment, moderate-quality evidence)[6,9,51,54,68]. Survivors treated with higher doses of C(S)RT (low-quality evidence)[32] and corticosteroids (moderate-quality evidence)[32,46,69] are at greater risk of low BMD. A dose threshold for increased risk of low BMD could not be determined from available literature for C(S)RT or corticosteroids. The effect of higher total body irradiation doses is unclear, as no studies investigated this. Furthermore, it is unknown whether administering dexamethasone leads to higher risk of low BMD than prednisone (no studies). However, very low-quality evidence for a greater risk of dexamethasone versus prednisone on lower BMD Z-scores as a continuum was found.[79] Survivors treated with

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abdominal/pelvic irradiation[6,9] also have increased risk of low BMD (low-quality evidence). No significant associations between low BMD and treatment with methotrexate (moderate-quality evidence),[6,9,54,68] ifosfamide (moderate-quality evidence),[6,9] cyclophosphamide (moderate-quality evidence),[6,9,68] and hematopoietic stem cell transplantation without TBI (low-quality evidence)[9,54] were found. Finally, the panel identified no studies that assessed the independent effect of cisplatin, 6-mercaptopurine, cyclosporine, tyrosine kinase inhibitors, or tacrolimus on low BMD in CAYA cancer survivors.

Other risk factors for low BMD

Survivors with GHD (moderate-quality evidence)[10,54,61,68,82,83] or hypogonadism (low-quality evidence)[10,38,51,54,61,68,82,84] have increased risk of low BMD. In more than half of the included studies, either some of the survivors received hormone replacement therapy or the percentage of treated survivors was not reported (Appendix pp. 321–323 and 326–328). Survivor characteristics and health behaviors associated with low BMD include low BMI (high-quality evidence),[6,9,40,42,51,54,56,64,68,69] male sex (high-quality evidence),[6,9,32,40–42,45,51,54,56,64,65,68] White race (moderate-quality evidence),40,41,51,65,68 lack of physical activity (moderate-quality evidence),[11,40,51,56] and current or prior smoking (moderate-quality evidence).[6,9,56,61] We found conflicting evidence for the association between alcohol consumption[61] and low BMD. No significant effect of age at cancer diagnosis[6,9,54,56,64,68,69] or inadequate dietary vitamin D and calcium intake (moderate-quality evidence, one study)[51] on the risk of low BMD was observed. However, we found an increased risk of low BMD for survivors with biochemical vitamin D deficiency (25OHD levels <20 ng/ml) (low-quality evidence).[65] None of the included studies assessed the risk of low BMD in relation to biochemical vitamin B deficiency, hyperthyroidism, or consumption of carbonated beverages.

Risk factors for fractures

Risk factors for fractures in CAYA cancer survivors include male sex (moderate-quality evidence)[41,58,81] and higher doses corticosteroids (low-quality evidence)[81] (Table 1). When evidence about other risk factors for fractures was available, it was of very low-quality or no increased risk was identified.

Diagnostic value of BMD surveillance modalities

No studies investigating the diagnostic value of quantitative computed tomography (QCT) compared to dual-energy X-ray absorptiometry (DXA) in CAYA cancer survivors were identified. However, two studies showed a significant correlation between QCT and DXA-derived BMD parameters in this population (low-quality evidence).[41,85] Very low-quality evidence indicated that the diagnostic value of quantitative ultrasound (QUS) compared to DXA was moderate.[86] No studies addressing the diagnostic value of QUS and peripheral QCT (pQCT) versus QCT, as well as the added value of QUS to QCT and DXA in predicting fractures in survivors, were identified. Also, the optimal site of BMD measurement has not been evaluated (no studies).

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Risk of (very) low BMD over time

The average time from diagnosis to the time of (very) low BMD development in survivors is unknown (no studies). Moderate-quality evidence suggests that BMD Z-scores increase from two years until at least ten years from completion of cancer treatment in CAYA cancer survivors.[32,40,49,64,71,87–90]

Interventions to maintain or increase BMD

Limited evidence for efficacy of interventions to remediate (very) low BMD in CAYA cancer survivors was identified. A significant effect of growth hormone replacement therapy on BMD in survivors with GHD was reported (very low-quality evidence).[91–93] However, no significant effect of weight-bearing physical exercise (low-quality evidence)[94] or calcium and vitamin D supplementation (very low-quality evidence)[43] on BMD was found. In addition, one study showed no significant effect of twice daily treatment with a vibrating plate on total body BMD Z-score in survivors in an intention-to-treat analysis, although there was significant improvement of tibial trabecular bone content among participants completing at least 70% of prescribed sessions.[95] The effects of bisphosphonates, parathyroid hormone, denosumab, vitamin B12 supplementation, and sex steroid replacement therapy were not studied in any of the included reports.

Potential benefits and harms of BMD surveillance

The potential benefits and harms of BMD surveillance are shown in Figure 2. Most importantly, it is unclear whether early treatment of BMD deficits leads to better skeletal health (i.e. no further progression, prevention of fractures) in CAYA cancer survivors, which is a prerequisite for BMD surveillance. However, in healthy children, weight-bearing physical exercise and calcium and vitamin D supplementation (in case of deficit) have been shown to maintain or enhance BMD.[96,97] Diagnosing (very) low BMD at an early stage among at-risk survivors could provide rationale for targeted healthy lifestyle recommendations. Furthermore, in older adults from the general population, treatment of (very) low BMD prevents fractures and consequent morbidity, decreased quality of life, and hospitalization.[98]

Translating evidence into recommendations

Who needs BMD surveillance?—The panel strongly recommends that CAYA cancer survivors and their healthcare providers should be aware of the risk of (very) low BMD and pay specific attention to possible consequences (e.g. acute and chronic back pain, low-trauma vertebral fractures [VF] and non-VF, and loss of height due to VF) after treatment with C(S)RT (high-quality evidence for very low BMD), TBI (high-quality evidence for low BMD), or corticosteroids (moderate-quality evidence for low BMD) (Figure 3).

Overall, the balance of desirable and undesirable anticipated effects of surveillance varies depending on the risk of (very) low BMD (Appendix pp. 366–373). In CAYA cancer survivors treated with C(S)RT (high-quality evidence for very low BMD), the panel was convinced that the potential benefits of BMD surveillance clearly outweigh the potential harms. For these survivors, we strongly recommend BMD surveillance. Mechanisms

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through which C(S)RT could cause very low BMD include hypogonadism, GHD, low BMI or lean mass, and obesity leading to diabetes.[10,99] These mechanisms may also apply to survivors treated with TBI, but evidence is more limited (high-quality evidence for low BMD, but no studies for very low BMD). Therefore, we made a moderate recommendation for BMD surveillance after treatment with TBI. A dose threshold for C(S)RT and TBI could not be determined, but the risk of (very) low BMD is likely associated with the risk of hypothalamic-pituitary axis, ovarian, or testicular injury after radiotherapy, and with age of radiotherapy administration. In addition, radiotherapy may impact bones directly and increase fracture risk, but this may not directly result from low BMD but from a perturbation in bone (re)modeling.[100]

It is well-known that corticosteroids (administered as anti-cancer treatment) have a detrimental effect on bone (especially trabecular bone) during its administration, resulting in an increased risk of low-trauma VF and non-VF.[101] However, for survivors with and without bone issues during therapy, the effect of corticosteroids on BMD more than two years after the last exposure is unclear, as longitudinal studies from corticosteroid initiation until many years after its termination are lacking. We found an increased risk of low BMD after corticosteroid treatment (moderate-quality evidence), but not of very low BMD (moderate-quality evidence) and fractures (very low-quality evidence). An increased risk of low BMD (moderate-quality evidence) and fractures (low-quality evidence) after higher doses corticosteroids was observed. The included longitudinal studies (mostly involving survivors treated with steroids) showed that BMD Z-scores increased after treatment cessation.[32,40,49,87–90] This observation is supported by literature from corticosteroid-treated adult populations, which shows that after corticosteroid termination, BMD increases and clinical fracture risk declines.[102,103] However, the age of corticosteroid administration may play a role in this respect, as corticosteroid exposure during puberty, a critical period for bone mass acquisition, may impact BMD more severely and permanently than during the prepubertal period.[104,105] In addition, the fracture risk associated with corticosteroids in other populations may be due to mechanisms not definitively assessed by BMD measurements, such as altered bone structure and increased risk of falls due to muscle weakness.[106] Ultimately, a large proportion of CAYA cancer survivors have been treated with (higher doses) corticosteroids, and it is unclear if the benefits of BMD surveillance outweigh the potential harms for this substantive group. Therefore, no recommendation can be formulated for or against BMD surveillance. This surveillance decision should be made by the survivor and healthcare provider together, after careful consideration of the potential harms and benefits (Figure 2 and Survivor Information Brochure) and additional risk factors (Figure 3).

Abdominal/pelvic irradiation was not included in the surveillance recommendations because the moderate-quality evidence for an increased risk of very low BMD was based on only one study, and because it is uncertain whether the risk of (very) low BMD for survivors treated with abdominal/pelvic irradiation who did not develop hypogonadism is increased.

In CAYA cancer survivors with hypogonadism or GHD, BMD measurements should be performed as part of standard endocrine care, which is best done by a medical bone health specialist in the context of hypogonadism or GHD management. A medical bone

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health specialist may include specialists such as an endocrinologist (most settings), internist, pediatrician, rheumatologist, or general practitioner, depending on country and setting. We refer to the IGHG guidelines premature ovarian insufficiency[107] and male gonadotoxicity[108] for survivors at risk of hypogonadism.

What surveillance modality should be used?—The included clinical practice guidelines in related populations all advise DXA (lumbar spine and total body less head [TBLH] or total hip, depending on age) for BMD surveillance.[24,109,110] The International Society of Clinical Densitometry (ISCD) 2019 pediatric position states that the lumbar spine and TBLH are the preferred sites for BMD measurements.[25] In addition, although hip measurements are generally not preferred in younger children due to variability in skeletal development and limited reference values, they can be useful in adolescents at-risk for bone fragility who would benefit from continuity of DXA measurements through transition into adulthood, as is the case for CAYA cancer survivors. One of the included guidelines recommends to avoid QCT for clinical application based on higher radiation doses applied compared to DXA (radiation exposure from pQCT is much lower, although slightly higher than DXA),[110] and the ISCD currently considers both quantitative modalities primarily research techniques.[24] Although the panel appreciates the value of volumetric BMD measured by (p)QCT, clear QCT-derived BMD thresholds associated with fractures have not yet been established. Furthermore, the panel has concerns about the availability of (p)QCT in most settings (expert opinion). We found that the diagnostic value of QUS compared to DXA is moderate in survivors (very-low quality evidence). However, QUS is not endorsed generally in diagnosing reduced BMD in CAYAs, and appropriate normative data are scarce (expert opinion). Based on these considerations, we recommend a DXA scan of the lumbar spine (posterior-anterior L1-L4), TBLH (in children and adolescents), and total hip (in adolescents and adults) for BMD surveillance (strong recommendation). QCT is currently not recommended for BMD surveillance (evidence-based guidelines and expert opinion). No recommendation for or against using pQCT and QUS for BMD surveillance was formulated.

When should surveillance be initiated and at what frequency should it be performed?—BMD Z-scores seem to increase in survivors from entry into long-term follow-up (LTFU) onwards. However, longitudinal studies that include more than two years off therapy investigating the average time to development of (very) low BMD to inform the timing of BMD surveillance were absent. In general, screening for a condition is only justified when results have treatment implications.[111] Bisphosphonates can effectively treat osteoporosis and reduce fracture risk in children and adults.[98,112] However, because bisphosphonates have several known and theoretical side-effects in children,[113] their use is presently only considered in individuals with severe bone fragility (fragility fractures ± very low BMD), and reduced potential for BMD restitution and vertebral body reshaping following VF.[104,114] Prophylactic bisphosphonate therapy (i.e. treating a low BMD Z-score in the absence of fractures) is not currently recommended.[110,114] This means that bisphosphonates are not indicated when BMD deficits are detected through primary surveillance in this age group. Initiating bisphosphonate treatment solely based upon very low BMD is also controversial in CAYA cancer survivors that have achieved PBM

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(around the age of 25 years), because of low absolute fracture risk, low bisphosphonate efficacy if active bone loss is absent, and potentially long treatment duration, but may be considered in older survivors under specific circumstances. PBM is predictive of bone status in later adulthood and therefore represents an important landmark in time.[14] However, the interval between entering LTFU and attaining the age of 25 years may be long for many survivors, which represents a period of potential increased risk of (very) low BMD and fractures. In addition, it is important to optimize PBM acquisition, and interventions such as exercise and hormone replacement therapy are most effective during puberty.[96,115] Hence, BMD surveillance is recommended at entry into LTFU (between two to five years following completion of therapy) and if normal (Z-score >−1), it is recommended to repeat surveillance at 25 years of age when PBM should be achieved (expert opinion). In the small proportion of survivors for whom these timepoints overlap or are in proximity, one DXA scan is sufficient. Between these two measurements and thereafter, BMD surveillance should be performed as clinically indicated based on BMD and ongoing risk assessment. Furthermore, BMD trajectories are more informative than a single, cross-sectional measurement.

What should be done when abnormalities are identified?—The two included clinical practice guidelines for BMD treatment state that bisphosphonate therapy should be reserved for patients with overt bone fragility.[109,110] Furthermore, these guidelines conclude that adequate dietary calcium and vitamin D intake is important for optimal bone health, and that supplementation is warranted in case of deficit. The presence of endocrinopathies such as hypogonadism and GHD should be evaluated and corrected. In addition, these guidelines recommend adequate weight-bearing physical activity, and one guideline underscores that negative lifestyles such as smoking and any alcohol use should be avoided.[110]

The panel drafted the following recommendations based on published evidence-based guidelines and expert opinion. In CAYA cancer survivors with a BMD Z-score ≤−2, referral to (or consultation of) a medical bone health specialist is recommended for further (endocrine) evaluation, interpretation of BMD findings, treatment, and follow-up (expert opinion). In survivors with a BMD Z-score ≤−1 and >−2, it is recommended to: 1) evaluate for the presence of endocrine defects (hypogonadism, GHD etc.), and consult a medical bone health specialist for further evaluation as clinically indicated (very low-quality evidence and evidence-based guidelines); and 2) repeat DXA after 2 years, and thereafter as clinically indicated based on BMD change (i.e. in case of BMD decline more than the DXA machine's least significant change[25]) and ongoing risk assessment (expert opinion). In all at-risk CAYA cancer survivors (i.e. those treated with C[S]RT, TBI, and/or corticosteroids), regardless of their BMD Z-score, it is recommended to counsel about the following lifestyle habits that are important to maintain or improve bone health: 1) engage in regular physical activity,[116] especially weight-bearing and falls prevention activities (evidence-based guidelines and expert opinion); 2) abstain from smoking (moderate-quality evidence and evidence-based guidelines) and limit or avoid alcohol intake (evidence-based guidelines); 3) consume adequate dietary vitamin D (at least 400 IU/day) and calcium (at least 500 mg/day) irrespective of vitamin D status, and advise vitamin D supplementation in survivors with 25OHD levels <20 ng/ml (plus calcium if the recommended amount of

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dietary calcium is not met) as per local or national guidelines (evidence-based guidelines and expert opinion); and 4) advise nutritional supplementation for survivors with low BMI or underweight (expert opinion).

Lastly, there were insufficient studies that assessed fracture outcomes to draw formal fracture-based surveillance recommendations. However, low-trauma VF are the clinical signature of osteoporosis, and low-trauma non-VF may also indicate a diagnosis of osteoporosis.[24] Therefore, it is reasonable to refer at-risk CAYA cancer survivors with a history of low-trauma VF and non-VF (from entry into LTFU onwards) to a medical bone health specialist for further examination and treatment.

Discussion

This study provides a comprehensive BMD surveillance strategy for CAYA cancer survivors, which could enhance early identification and adequate treatment and follow-up of survivors with (very) low BMD in a variety of LTFU settings, with the goal to prevent clinically relevant fractures and their consequences. In addition, the guideline panel identified gaps in the current literature that could guide future research to improve BMD surveillance and fracture prevention strategies in survivors (Panel).

(Very) low BMD in survivors results from a complex, multifactorial process. We identified C(S)RT, abdominal/pelvic irradiation, total body irradiation, and corticosteroids as the main treatment-related risk factors for (very) low BMD. The primary mechanism through which these treatment modalities impact BMD varies; however, they can all lead to primary or secondary hypogonadism.[117] Hypogonadism is a well-established cause of osteoporosis in the general population,[118,119] and was independently associated with (very) low BMD in this review. Therefore, we hypothesize that hypogonadism is also a key driver of (very) low BMD in CAYA cancer survivors. Interestingly, moderate-quality evidence suggests no increased risk of (very) low BMD after treatment with ifosfamide or cyclophosphamide, two alkylating agents that may induce hypogonadism, especially at higher doses.[107] We propose that the number of survivors with hypogonadism following ifosfamide or cyclophosphamide at any dose may have been too small (and/or received sex steroid replacement therapy) in available studies to detect a significant association with (very) low BMD.

As in the general population, host and lifestyle factors (e.g. White race, low BMI, smoking) contribute to the risk of (very) low BMD in CAYA cancer survivors. This makes it difficult to designate a distinct group of survivors at highest absolute risk of (very) low BMD solely based on prior cancer treatment. In this guideline, we recommend BMD surveillance for those survivors with an excess risk of very low BMD based on their prior cancer treatment (i.e. those treated with C[S]RT or TBI). Another approach is to incorporate treatment-related, demographic, and lifestyle factors into a prediction model for the absolute risk of (very) low BMD, which has been done for White adult childhood cancer survivors.[6] However, this model needs to be further validated in survivors younger than 18 and older than 40 years of age, as well as in non-White survivors, before its use can generally be recommended in CAYA cancer survivors.

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We recommend using DXA as the optimal modality for BMD surveillance in CAYA cancer survivors. However, in growing children more so than in adults, several limitations should be considered when interpreting areal BMD Z-scores generated by DXA. First, areal BMD may be underestimated in shorter individuals (for example as a result of GHD or delayed pubertal development) due to the confounding effect of bone size.[120] To correctly interpret BMD findings in these individuals, the bone mineral apparent density (BMAD, g/cm^3) can be estimated to reduce this confounding effect,[121,122] and a medical bone health specialist can be consulted to further assist in BM(A)D interpretation. Second, although the association between lumbar spine BMD and VF remains consistent, Z-scores vary significantly in children depending upon the reference database that is used.[123] This is further complicated by the fact that a Z-score of −1.5 for example might be normal for one individual, but abnormal for another, and that fragility fractures can occur at a BMD Z-score better than −2.[123] Monitoring BMD trajectory by performing serial DXA scans (recommended in this guideline) could overcome some of these limitations.

An important aspect of the IGHG guideline process is the identification of gaps in knowledge and development of directions for future research. According to our findings, future studies should investigate the risk of reduced BMD after several treatment modalities and disease types (such as bone and soft tissue sarcomas) in more detail, and further develop and validate prediction models for very low BMD and low-trauma fractures (Panel). One important development in recent years is that bone research has expanded to consider not only BMD, but also bone structure, as an important indicator of bone strength. For example, it is now understood that low-trauma VF are a key sign of osteoporosis in the cancer setting, but that such fractures are frequently asymptomatic and thereby go undetected in the absence of VF imaging. While a longitudinal study of incident low-trauma VF and non-VF has been carried out up to six years following childhood acute lymphoblastic leukemia diagnosis,[101] studies beyond this duration and including other cancer settings are presently lacking. Therefore, additional longitudinal studies are needed to identify the risk and risk factors of incident low-trauma fractures, the most frequent sites of fractures, as well as the BMD trajectory with age.

Several limitations of the available literature are important in interpreting our BMD surveillance recommendations. For instance, much literature is available on the risk of BMD Z-scores ≤−1, but less is known about the risk of BMD Z-scores ≤−2 and low-trauma fractures, which are more clinically relevant outcomes. Furthermore, only few of the available studies discussed the bone health of patients with sarcomas, and the follow-up times differed significantly across included studies. In addition, although all studies performed multivariable analyses, some failed to adjust for all essential confounders (e.g., sex, BMI, age, and Tanner stage). These limitations hinder comparison of results between studies. Finally, recommendations regarding diagnostic modality, optimal timing of BMD surveillance, and interventions for survivors with reduced BMD were extrapolated from non-cancer populations and clinical expertise due to lack of evidence in CAYA cancer survivors.

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Conclusion

This IGHG guideline provides harmonized recommendations for BMD surveillance that may improve health outcomes by facilitating more consistent LTFU care for current CAYA cancer survivors. In addition, it promotes strategically planned ongoing research that will inform future guideline updates.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgements

We have no role of funding source to disclose. All authors had full access to the full data in the study and accept responsibility to submit for publication. We thank Prof. Charles Sklar (Memorial Sloan Kettering Cancer Center, New York, NY, USA) and Prof. Jean-Marc Kaufman (Ghent University Hospital, Ghent, Belgium) for critically appraising the recommendations and manuscript as external reviewers and Debbie Crom (St. Jude Children's Research Hospital, Memphis, TN, USA), Lisa Goerens (CCI Europe, Luxembourg), Marie Barth (CCI Europe, Germany), Tiago Costa (CCI Europe, Portugal), Mihael Severinac (CCI Europe, Croatia), and Zuzana Tomášiková (CCI Europe, Switzerland) as survivor representatives.

Declaration of interest

LMW received grants and personal fees from Novartis and Amgen (with funds to the Children's Hospital of Eastern Ontario Research Institute), outside the submitted work. KKN received grants from the National Institutes of Health and the American Lebanese Syrian Associated Charities (ALSAC), during the conduct of the study. All other authors declare no competing interests.

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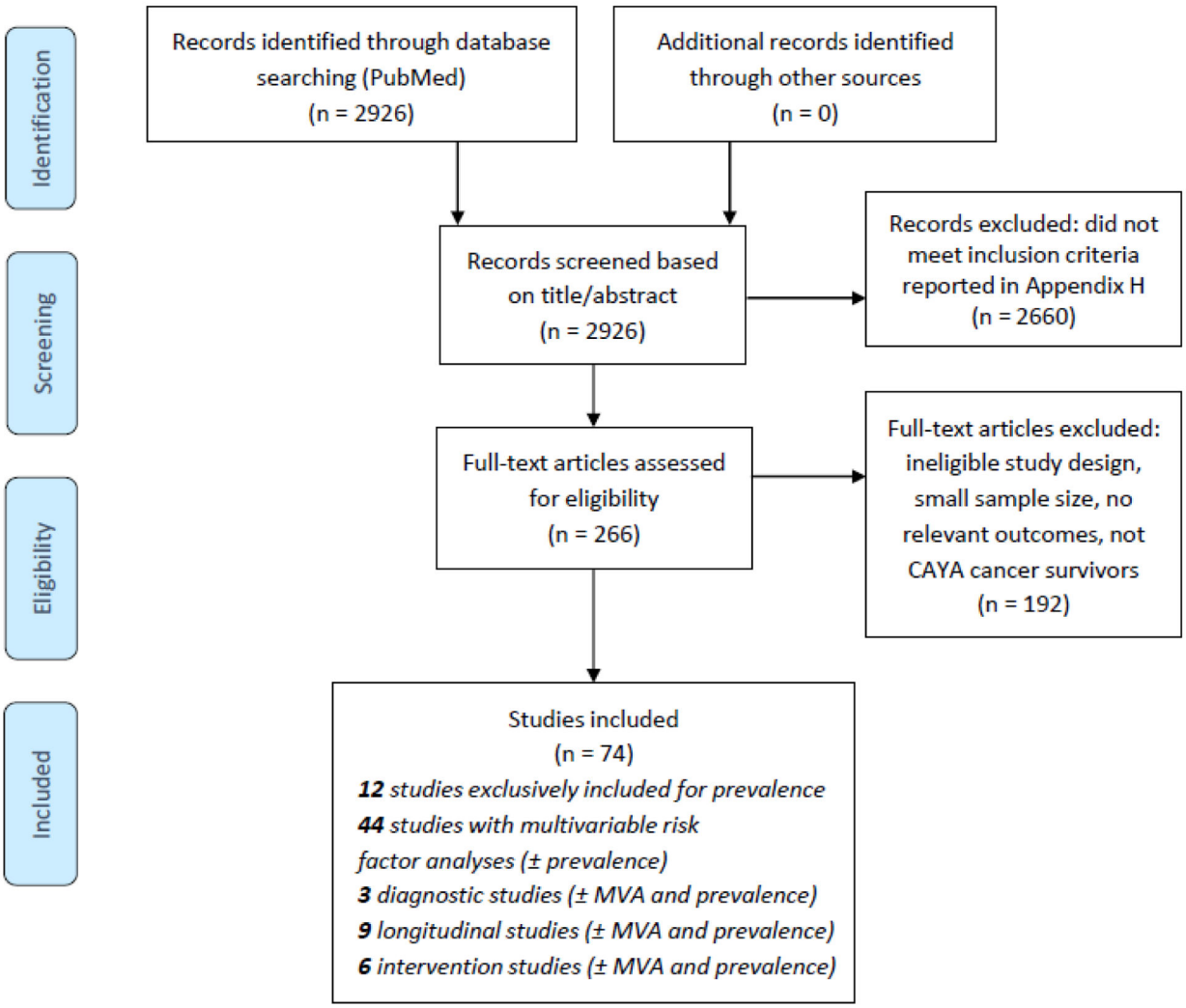

Figure 1.

PRISMA flow chart of selected studies.

Abbreviations: CAYA=childhood, adolescent, and young adult; MVA=multivariable analyses.

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Potential benefits:

- DXA can accurately detect low (Z-score ≤-1) and very low (Z-score ≤-2) BMD (gold standard) (*evidence-based guidelines*[24,109,110]).
- Optimization of bone health could prevent further BMD Z-score decline (*evidence-based guidelines*[109,110]*):*
  - Supplementation of calcium and vitamin D
  - Avoidance of negative lifestyles such as smoking and alcohol use
  - Participation in recommended (weight-bearing) physical activity
  - Evaluation of endocrine defects such as hypogonadism, and hormone replacement therapy
- Referral to a medical bone health specialist may be beneficial for further evaluation, interpretation, treatment, and follow-up. Further treatment with osteoporosis medication (e.g. bisphosphonates) may be warranted in survivors with severe bone fragility and low potential for BMD restitution and vertebral body reshaping (*expert opinion*).
- In older adults from the general population, treatment of low and very low BMD prevents fractures and consequent morbidity, decreased quality of life, and hospitalization,[98] although in CAYA cancer survivors, evidence is lacking to support these benefits (*evidence-based guidelines*[109,110]).
- Diagnosing reduced BMD at an early stage could provide rationale for more targeted healthy lifestyle recommendations, and certain interventions could be initiated at the time they may be most effective (before the end of puberty).[96,115] This may reduce the need for treatment with bone-modifying agents such as bisphosphonates that may cause considerable side-effects (*expert opinion*).
- CAYA cancer survivors who do not have low or very low BMD when they undergo surveillance may benefit from being reassured that their fracture risk is less likely to be increased (*expert opinion*), although they should appreciate that BMD is only one parameter of bone strength.
- The prevalence of frailty (accelerated aging) is increased in CAYA cancer survivors compared to siblings,[124] and therefore, they could be at higher risk of falls (and consequent fractures). For this reason, identifying (very) low BMD in this group at an early stage could be of particular importance. Falls prevention methods could be recommended for these survivors (*expert opinion*).

Potential harms:

- It is unclear whether early treatment of low or very low BMD leads to better health (i.e. no further progression, prevention of fractures) in CAYA cancer survivors (*no studies*).
- BMD surveillance may increase the risk of misdiagnosis, overdiagnosis, and overtreatment (i.e. incorrect diagnosis of (very) low BMD, or identification of (very) low BMD that would never lead to a fracture, possibly resulting in unnecessary treatment with, for example, bisphosphonates, which may lead to side-effects) (*expert opinion*).
- BMD surveillance by DXA may lead to potential harms from radiation exposure (especially in the context of cumulative radiation dose after treatment for CAYA cancer), although the dose of one DXA scan is considered negligible (less than one chest X-ray or a short flight[125]) (*expert opinion*).
- BMD surveillance may cause stress and anxiety (although the stress and anxiety generally resolve post screening) and the self-perception of being a patient versus healthy (*expert opinion*).

- CAYA cancer survivors might be falsely reassured that they do not have an increased fracture risk when their BMD is normal (*expert opinion*).
- DXA may be costly.

Figure 2.

Potential benefits and harms of bone mineral density surveillance for childhood, adolescent and young adult cancer survivors.

Abbreviations: BMD=bone mineral density; CAYA=childhood, adolescent and young adult; DXA=dual-energy X-ray absorptiometry.

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General recommendation
CAYA cancer survivors and their healthcare providers should be aware of the risk of low (Z-score ≤-1 and >-2) and very low (Z-score ≤-2) bone mineral density, and pay specific attention to possible consequences (e.g. acute and chronic back pain, vertebral and non-vertebral low-trauma fractures, and loss of height due to vertebral fractures) after treatment with: • Cranial or craniospinal radiotherapy (high-quality evidence for very low BMD) • Total body irradiation (high-quality evidence for low BMD, unknown effect for very low BMD) • Corticosteroids as anti-cancer treatment (moderate-quality evidence for low BMD, no significant effect for very low BMD) Other risk factors for low and very low bone mineral density in CAYA cancer survivors include[1]: • Hypogonadism (moderate-quality evidence for very low BMD; BMD assessment is recommended according to standard endocrine care, which is best done by a medical bone health specialist[2]) • Growth hormone deficiency (moderate-quality evidence for low BMD; BMD assessment is recommended according to standard endocrine care, which is best done by a medical bone health specialist[2]) • Low BMI or underweight (high-quality evidence for very low BMD) • Male sex (moderate-quality evidence for very low BMD) • White race (moderate-quality evidence for low BMD) • Lack of physical activity[3] (moderate-quality evidence for low BMD) • Current or prior smoking (moderate-quality evidence for low BMD)
Who needs bone mineral density surveillance?
Bone mineral density surveillance is recommended for CAYA cancer survivors treated with cranial or craniospinal radiotherapy (high-quality evidence for very low BMD).
Bone mineral density surveillance is reasonable for CAYA cancer survivors treated with TBI (high-quality evidence for low BMD).
Due to insufficient evidence[4], no recommendation can be formulated for or against BMD surveillance for CAYA cancer survivors treated with corticosteroids as anti-cancer treatment. The surveillance decision should be made by the CAYA cancer survivor and healthcare provider together, after careful consideration of the potential harms and benefits (see Survivor Information Brochure) and additional risk factors.
What surveillance modality should be used?
A DXA scan of the lumbar spine (posterior-anterior L1-L4), total body less head (in children and adolescents), and total hip (in adolescents and adults) are recommended for surveillance of bone mineral density (evidence-based guidelines).
QCT is not recommended for surveillance of bone mineral density (evidence-based guidelines and expert opinion).
When should surveillance be initiated and at what frequency should it be performed?
BMD surveillance is recommended at entry into LTFU (between two to five years following completion of therapy), and if normal (Z-score >-1), it is recommended to repeat surveillance at 25 years of age when peak bone mass should be achieved. Between these two measurements and thereafter, BMD surveillance should be performed as clinically indicated based on BMD and ongoing risk assessment (expert opinion).
What should be done when abnormalities are identified?
In CAYA cancer survivors with a BMD Z-score ≤-2, referral to (or consultation of) a medical bone health specialist[2] is recommended for further (endocrine) evaluation, interpretation of BMD findings, treatment, and follow-up (expert opinion).
In CAYA cancer survivors with a BMD Z-score ≤-1 and >-2, it is recommended to: • Evaluate for the presence of endocrine defects (hypogonadism, GHD etc.), and consult a medical bone health specialist[2] for further evaluation and interpretation of BMD findings as clinically indicated (very low -quality

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evidence and evidence-based guidelines)

- Repeat DXA after 2 years, and thereafter as clinically indicated based on BMD change (i.e. in case of BMD decline more than the DXA machine's least significant change) and ongoing risk assessment (expert opinion)

In all at-risk CAYA cancer survivors[5], regardless of their BMD Z-score, it is recommended to counsel about lifestyle habits that are important to maintain or improve bone health:

- Engage in regular physical activity[3], especially weight-bearing and fall prevention activities (evidence-based guidelines and expert opinion)
- Abstain from smoking (moderate-quality evidence for low BMD and evidence-based guidelines)
- Limit or avoid alcohol intake (evidence-based guidelines)
- Consume adequate dietary vitamin D (at least 400 IU/day) and calcium (at least 500 mg/day) irrespective of vitamin D status, and advise vitamin D supplementation in survivors with 25OHD levels <20 ng/ml[6] (plus calcium if the recommended amount of dietary calcium is not met) as per local or national guidelines (evidence-based guidelines and expert opinion)
- Advise nutritional supplementation for CAYA cancer survivors with low BMI or underweight (expert opinion)

It is reasonable to refer at-risk CAYA cancer survivors[5] with a history of low-trauma vertebral and non-vertebral fractures (from entry into LTFU onwards) to a medical bone health specialist[2] for further evaluation and treatment (expert opinion).

Figure 3.

Harmonized recommendations for bone mineral density surveillance for CAYA cancer survivors.

Abbreviations: BMD=bone mineral density; BMI=body mass index; CAYA=childhood, adolescent and young adult; DXA=dual energy X-ray absorptiometry; LTFU=long-term follow-up; PBM=peak bone mass; TBI=total body irradiation.

[1]As in the general population (except for sex; female sex in the general population); [2]A medical bone health specialist is defined as any specialist who is caring for BMD deficits in CAYA cancer survivors, such as an endocrinologist (most settings), internist, pediatrician, rheumatologist, family physician, or general practitioner, depending on country and setting; [3]The WHO global recommendation on physical activity for health for adults is 150 minutes of moderate-intensity activity (or equivalent) per week, measured as a composite of physical activity undertaken across multiple domains: for work (paid and unpaid, including domestic work); for travel (walking and cycling); and for recreation (including sports). For adolescents, the recommendation is 60 minutes of moderate- to vigorous-intensity activity daily; [4]Insufficient evidence to determine if early detection of low BMD after treatment with corticosteroids reduces morbidity in CAYA cancer survivors, and whether the risk of very low BMD is increased in the long-term; [5]Survivors treated with C(S)RT (high-quality evidence), TBI (high-quality evidence), or corticosteroids (moderate-quality evidence); [6]Target 25OHD levels should be >20 ng/ml.

Green representing a strong recommendation to do with a low degree of uncertainty; Yellow representing a moderate recommendation to do with a higher degree of uncertainty; Red representing a recommendation not to do.

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Why should I be aware of the risk of low bone mineral density (weak bones)?

- Bone mineral density is an important determinant of bone strength. This means that if you have low bone mineral density (weak bones), you probably break your bones more easily.
- Having weak bones around the age of 25 (when your bones should be the heaviest) predicts for osteoporosis and bone fractures later in life.
- As a survivor of childhood, adolescent or young adult cancer you may have a higher risk of developing weak bones compared to people of similar age in the general population.
- If your brain and spinal cord were exposed to radiation as part of your treatment for a childhood, adolescent, or young adult cancer (cranial[spinal] irradiation), or if you were treated with total body irradiation, you have an increased risk of developing weak bones.
- If you were treated with corticosteroids (as anti-cancer treatment) you may have an increased risk of weak bones as well. However, it is unclear if corticosteroids can lead to weak bones in the long term.
- While some people treated with cranial(spinal) irradiation, total body irradiation, and/or corticosteroids will develop weak bones at a young age, most will not.
- However, among those who develop weak bones, detecting it early can possibly prevent bone fractures and may therefore reduce consequences such as pain, surgery, and temporary immobilization.
- It is possible to detect weak bones early by having bone mineral density screening, but bone mineral density screening has advantages and disadvantages.
- This information sheet can be used to help you and your healthcare provider decide if having bone mineral density screening is the right choice for you.

What is bone mineral density screening?

- Bone mineral density screening is performed with a bone scan that uses low dose X-rays to see how strong your bones are.

What are the potential advantages of having bone mineral density screening?

- You may feel reassured if you have normal bone mineral density at this time. However, weak bones may still develop in the future, and your fracture risk may still be increased due to other reasons.
- Early detection would allow doctors to monitor the bone mineral density course over time. In addition, early detection would allow referral to a specialized bone doctor who can further evaluate your bone health, which may both help to determine if/when treatment is needed.
- You may be more likely to have weak bones detected at an earlier timepoint when certain interventions may be most effective (before the end of puberty), and as a result, bone fractures may be prevented.

What are the potential disadvantages of having bone mineral density screening?

- You may experience anxiety and stress about having bone mineral density screening and what the test results will show.
- You may feel more like a patient rather than a healthy survivor if you decide to have bone mineral density screening.

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- You may be incorrectly diagnosed with weak bones (misdiagnosis), or diagnosed with weak bones that never would have caused fractures (overdiagnosis), although your doctor carefully considers treatment.
- We do not know if early treatment of weak bones leads to better health (no further weakening of the bones or prevention of fractures) in childhood, adolescent and young adult cancer survivors. However, in the general population, we know that this is the case.
- The diagnosis of weak bones may affect your ability to obtain healthcare and/or life insurance.

What are the potential disadvantages associated with this bone scan?

- This bone scan is associated with potential harms from radiation exposure (especially in the context of cumulative radiation dose after cancer treatment), although the dose of one scan is considered negligible (less than one chest X-ray or a short flight).
- This bone scan may be costly and may not be covered by your health insurance. However, your healthcare provider could write a letter of medical necessity to explain that you are at increased risk of weak bones and why you may benefit from a bone scan.

What are the international screening recommendations?

- If you were treated with radiotherapy to your brain or spinal cord, total body irradiation, and/or corticosteroids, it is important that you are aware of the risk of weak bones, and pay specific attention to their possible consequences (acute back pain, [spinal] fractures, and loss of height due to spinal fractures).
- If you were treated with radiotherapy to your brain or spinal cord, bone mineral density screening is recommended at entry into long-term follow-up (beginning two or more years following completion of therapy) and at 25 years of age.
- If you were treated with total body irradiation, bone mineral density screening is reasonable at entry into long-term follow-up and at 25 years of age.
- If you were treated with corticosteroids as anti-cancer treatment, we cannot recommend for or against routine bone mineral density screening because we do not know if your health outcomes will be better if we detect weak bones early. It is important that you make the decision whether or not to screen together with your healthcare providers, oncology and survivorship team, and individual support networks. Careful consideration of the potential advantages and disadvantages is advised.

Thank you for taking the time to read this information sheet. If you have any questions regarding the information included in this brochure or if you require emotional support and advice regarding your thoughts and feelings, please contact your healthcare provider for advice and support.

Figure 4.

Potential advantages and disadvantages of bone mineral density surveillance for childhood, adolescent and young adult cancer survivors – A Survivor Information Brochure.

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Table 1.

Conclusions and quality of the evidence for bone mineral density surveillance in CAYA cancer survivors.

Who needs bone mineral density surveillance?				
Risk and risk factors for low BMD, very low BMD, lower BMD Z-score, and fractures in CAYA cancer survivors diagnosed up to 25 years of age				
	Very low BMD (Z-score ≤−2)	**Low BMD (Z-score ≤−1 and ≤−2)**	**Lower BMD Z-score (continuous)**	**Fractures (all types)**
Risk				
Risk	↑⊕⊕⊕⊖ MODERATE[6,7,9,10,30–37,39–44,47–55,58–62,64,66–71]	↑⊕⊕⊕⊖ MODERATE[6,7,9–11,30–70]	↑⊕⊕⊕⊖ MODERATE[30,31,33,38,39,41,43,44,50,52–55,58,60,61,64,66,67,69–79]	↑⊕⊖⊖⊖ VERY LOW[7,8,80]
Risk after low BMD/fracture	No studies	No studies	No studies	No studies
Host factors				
Male sex	↑⊕⊕⊕⊖ MODERATE[6,41,54]	↑⊕⊕⊕⊕ HIGH[6,9,64,65,68,32,40–42,45,51,54,56]	↑⊕⊕⊖⊖ LOW[9,35,41,43,44,49,53,67,74,76]	↑⊕⊕⊕⊖ MODERATE[41,58,81]
Age at diagnosis	=⊕⊕⊕⊕ HIGH[6,54,69]	=⊕⊕⊖⊖ LOW[6,9,54,56,64,68,69]	⇕⊕⊖⊖⊖ VERY LOW[9,35,44,53,67,70,72]	=⊕⊕⊖⊖ LOW[80,81]
White race	↑⊕⊖⊖⊖ VERY LOW[41]	↑⊕⊕⊕⊖ MODERATE[40,41,51,65,68]	↑⊕⊕⊕⊖ MODERATE[41,43]	↑⊕⊖⊖⊖ VERY LOW[80]
Low BMI/weight/ lean mass	↑⊕⊕⊕⊕ HIGH[6,54,69]	↑⊕⊕⊕⊕ HIGH[6,9,40,42,51,54,56,64,68,69]	↑⊕⊕⊕⊖ MODERATE[9,33,35,36,43,52,53,72,74,126]	No studies
Certain SNPs	=⊕⊖⊖⊖ VERY LOW[66]	=⊕⊖⊖⊖ VERY LOW[66]	↑⊕⊕⊖⊖ LOW[9,73,74,78]	↑⊕⊖⊖⊖ VERY LOW[127]
Family history of OP/#	No studies	No studies	No studies	No studies
Treatment factors				
Corticosteroids (y/n)	=⊕⊕⊕⊖ MODERATE[6,54]	↑⊕⊕⊕⊖ MODERATE[6,9,51,54,68]	↑⊕⊖⊖⊖ VERY LOW[9,38,44]	=⊕⊖⊖⊖ VERY LOW[80,127]
Higher corticosteroid dose	↑⊕⊖⊖⊖ VERY LOW[69]	↑⊕⊕⊕⊖ MODERATE[32,46,69]	↑⊕⊕⊕⊖ MODERATE[41,67,70,72,74]	↑⊕⊕⊖⊖ LOW[81]
DEXA vs. PRED	No studies	No studies	=⊕⊖⊖⊖ VERY LOW[79]	No studies
Methotrexate (y/n)	=⊕⊕⊕⊖ MODERATE[6,54]	=⊕⊕⊕⊖ MODERATE[6,9,54,68]	=⊕⊕⊖⊖ LOW[9,38]	↑⊕⊖⊖⊖ VERY LOW[80,81]
Higher methotrexate dose	No studies	=⊕⊕⊖⊖ LOW[32]	=⊕⊖⊖⊖ VERY LOW[70]	↑⊕⊖⊖⊖ VERY LOW[127]
Ifosfamide (y/n)	=⊕⊕⊕⊖ MODERATE[6]	=⊕⊕⊕⊖ MODERATE[6,9]	=⊕⊕⊖⊖ LOW[9]	=⊕⊖⊖⊖ VERY LOW[80]

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Higher ifosfamide dose	No studies	No studies	No studies	No studies
Cyclophosphamide (y/n)	⊕⊕⊕⊖ MODERATE[6]	⊕⊕⊕⊖ MODERATE[6,9,68]	⊕⊕⊖⊖ LOW[9]	⊕⊖⊖⊖ VERY LOW[80]
Higher cyclo dose	No studies	No studies	⊕⊕⊖⊖ LOW[68,70]	No studies
Cisplatin (y/n)	No studies	No studies	No studies	No studies
Higher cisplatin dose	No studies	No studies	No studies	No studies
6-MP (y/n)	No studies	No studies	No studies	No studies
Higher 6-MP dose	No studies	No studies	⊕⊖⊖⊖ VERY LOW[70]	No studies
Cyclosporine (y/n)	No studies	No studies	No studies	No studies
Higher cyclosporine dose	No studies	No studies	No studies	No studies
TKIs (y/n)	No studies	No studies	No studies	No studies
TKI dose	No studies	No studies	No studies	No studies
Tacrolimus (y/n)	No studies	No studies	No studies	No studies
Higher tacrolimus dose	No studies	No studies	No studies	No studies
C(S)RT (y/n)	⬆⊕⊕⊕⊕ HIGH[6,54]	⬆⊕⊕⊕⊕ HIGH[6,9,32,38,51,54,64,68]	⬆⊕⊕⊖⊖ LOW[9,33,38,50,52,67,72,79]	⊕⊕⊖⊖ LOW[81]
Higher C(S)RT dose	No studies	⬆⊕⊕⊖⊖ LOW[32]	No studies	⊕⊖⊖⊖ VERY LOW[127]
HSCT (y/n)	⊕⊕⊖⊖ LOW[54]	⊕⊕⊖⊖ LOW[9,54]	⬆⊕⊖⊖ VERY LOW[9,44]	No studies
TBI (y/n)	No studies	⬆⊕⊕⊕⊕ HIGH[9,45,54,64,65]	⬆⊕⊕⊖⊖ LOW[9,44,67,75]	No studies
Higher TBI dose	No studies	No studies	No studies	No studies
Abdominal/pelvic RT (y/n)	⬆⊕⊕⊕⊖ MODERATE[6]	⬆⊕⊕⊖⊖ LOW[6,9]	⊕⊕⊖⊖ LOW[9]	⊕⊖⊖⊖ VERY LOW[80]
Higher abd./pelvic RT dose	No studies	No studies	No studies	No studies
Medical conditions				
GHD (y/n)	⬆⊕⊕⊖⊖ LOW[10,54,82,83]	⬆⊕⊕⊕⊖ MODERATE[10,54,61,68,82,83]	⊕⊕⊖⊖ LOW[67,75]	No studies
Hypogonadism (y/n)	⬆⊕⊕⊕⊖ MODERATE[10,54]	⬆⊕⊕⊖⊖ LOW[10,38,51,54,61,68,82,84]	⬆⊕⊖⊖ VERY LOW[38,67]	No studies

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Vitamin D deficiency (y/n)	No studies	No studies	No studies	No studies
Hyperthyroidism (y/n)	No studies	No studies	No studies	No studies
Endocrine dysfunction * (y/n)	No studies	↑⊕⊖⊖⊖ VERY LOW[62]	No studies	No studies
Health behaviors				
Inadequate vit. D intake (y/n)	No studies	=⊕⊕⊕⊖ MODERATE[51]	No studies	No studies
Vitamin D deficiency (y/n)	No studies	↑⊕⊕⊖⊖ LOW[65]	No studies	No studies
Inadequate Ca intake (y/n)	No studies	=⊕⊕⊕⊖ MODERATE[51]	↑⊕⊖⊖⊖ VERY LOW[33]	No studies
Inadequate vit. B intake (y/n)	No studies	No studies	No studies	No studies
Lack of exercise (y/n)	No studies	↑⊕⊕⊕⊖ MODERATE[11,40,51,56]	↑⊕⊕⊖⊖ LOW[39,63]	=⊕⊕⊖⊖ LOW[11,80]
Current/prior smoking (y/n)	=⊕⊕⊕⊖ MODERATE[6]	↑⊕⊕⊕⊖ MODERATE[6,9,56,61]	=⊕⊕⊖⊖ LOW[9]	↑⊕⊖⊖⊖ VERY LOW[80]
Alcohol consumption (y/n)	No studies	⇕⊕⊖⊖⊖ VERY LOW[61]	No studies	No studies
Carbonated beverages (y/n)	No studies	No studies	No studies	No studies
What surveillance modality should be used?				
Diagnostic value to detect (very) low BMD in CAYA cancer survivors diagnosed up to 25 years of age				
Variable			**Outcome**	**Quality of evidence**
Diagnostic value of *QCT* vs. DXA			Unknown	No studies
Correlation between *QCT and DXA derived BM(A)D and BMD Z-scores*			Significant (r 0.33–0.64)	⊕⊕⊖⊖ LOW[41,85]
Diagnostic value of *QUS* vs. DXA			Moderate	⊕⊖⊖⊖ VERY LOW[86]
Diagnostic value of *QUS* vs. QCT			Unknown	No studies
Diagnostic value of *pQCT* vs. QCT			Unknown	No studies
Added value of *QUS* to QCT and DXA in predicting fractures			Unknown	No studies
Location of BMD measurement (lumbar spine, total body and/or hip) that should be evaluated			Unknown	No studies
When should surveillance be initiated and at what frequency should it be performed?				
Risk over time of (very) low BMD in CAYA cancer survivors diagnosed up to 25 years of age				

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Variable	Outcome	Quality of evidence
Course of BMD Z-scores over time from 2 years until at least 10 years since end of cancer treatment	Increase	⊕⊕⊕⊖ MODERATE[32,40,49,64,71,87–90]
Latency time of low BMD and fractures	Unknown	No studies
Risk of fractures for *low BMD* vs. normal BMD	Increased	⊕⊕⊖⊖ LOW[58,68]
Risk of fractures for *lower BMD* vs. higher BMD	Not significant	⊕⊕⊖⊖ LOW[81]
What should be done when abnormalities are identified?		
Use of medical interventions to improve BMD in CAYA cancer survivors diagnosed up to 25 years of age		
Variable	**Outcome**	**Quality of evidence**
Effect of *growth hormone replacement therapy* in GH deficient survivors	Significant	⊕⊖⊖⊖ VERY LOW[91–93]
Effect of *calcium and vitamin D supplementation*	Not significant	⊕⊖⊖⊖ VERY LOW[43]
Effect of *weight-bearing physical exercise*	Not significant	⊕⊖⊖⊖ VERY LOW[94]
Effect of *twice daily treatment with a vibrating plate*	Not significant (intention-to-treat analysis) Significant (per-protocol analysis)	⊕⊖⊖⊖ VERY LOW[95]
Effect of *bisphosphonates*	Unknown	No studies
Effect of *PTH*	Unknown	No studies
Effect of *Denosumab*	Unknown	No studies
Effect of *vitamin B12 supplementation*	Unknown	No studies
Effect of *sex hormone replacement therapy*	Unknown	No studies

*GHD, hypogonadism or thyroid dysfunction.

⬆ indicates an increased risk, = indicates no significant effect, and ⇳ indicates conflicting evidence.

Abbreviations: BMD=bone mineral density; BMI=body mass index; CAYA=childhood, adolescent, and young adult; CRT=cranial irradiation; CSRT=craniospinal irradiation; DEXA=dexamethasone; DXA=dual-energy X-ray absorptiometry; GH=growth hormone; GHD=growth hormone deficiency; HSCT=hematopoietic stem cell transplantation; OP=osteoporosis; PRED=prednisone; PTH=parathyroid hormone; pQCT=peripheral quantitative computed tomography; QCT=quantitative computed tomography; QUS=quantitative ultrasound; RT=radiotherapy; SNP=single nucleotide polymorphism; TBI=total body irradiation; TKI=tyrosine kinase inhibitors; y/n=yes/no; 6-MP=6-mercaptopurine; #=fracture.

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Panel.

Gaps in knowledge of BMD deficits and fractures in childhood, adolescent, and young adult cancer survivors and directions for future research.

Effect of different types of abdominal/pelvic irradiation (independent of hypogonadism) on the risk of low and very low BMD in female CAYA cancer survivors.
Effect of corticosteroids on the risk of low and very low BMD in CAYA cancer survivors with increasing follow-up time.
Effect of the age of corticosteroid treatment on the risk of low and very low BMD.
Independent effect of TBI and HSCT on the risk of low and very low BMD in CAYA cancer survivors.
Safe corticosteroid dose with regard to the risk of low and very low BMD in CAYA cancer survivors, and if so, what is this dose.
Safe C(S)RT dose with regard to the risk of low and very low BMD in CAYA cancer survivors, and if so, what is this dose.
Safe TBI dose with regard to the risk of low and very low BMD in CAYA cancer survivors, and if so, what is this dose.
Risk and risk factors of low and very low BMD in CAYA cancer survivors older than 40 years.
Sex- and pubertal stage-based differences in risk factors for low and very low BMD in CAYA cancer survivors.
Risk and risk factors of low and very low BMD in CAYA cancer survivors treated for bone or soft tissue sarcomas.
Risk and risk factors of incident low-trauma vertebral and non-vertebral fractures in CAYA cancer survivors, including treatment-related risk factors and other risk factors such as very low BMD, history of fractures, and maternal hip fracture etc. (included in the FRAX® fracture risk profile for older adults), the most frequent sites of fractures, and the disability and impact on quality of l ife resulting from low-trauma fractures.
Further improvement and validation of prediction models (including demographic, lifestyle, and treatment factors) for low and very low BMD, and development of a prediction model for low-trauma fractures in CAYA cancer survivors.
Risk and risk factors of impaired bone structure and its association (± BMD) with low-trauma fractures in CAYA cancer survivors.
BMD trajectory and latency time of low-trauma fractures from cancer diagnosis into very long-term follow-up.
Association between QCT, pQCT and QUS measurements and fracture risk in CAYA cancer survivors.

Abbreviations: BMD=bone mineral density; CAYA=childhood, adolescent, and young adult; C(S)RT=cranio(spinal) radiotherapy; HSCT=hematopoietic stem cell transplantation; (p)QCT=(peripheral) quantitative computed tomography; TBI=total body irradiation.